

Energy Harvesting Circuits and Systems

EE4C10 track-specific lecture

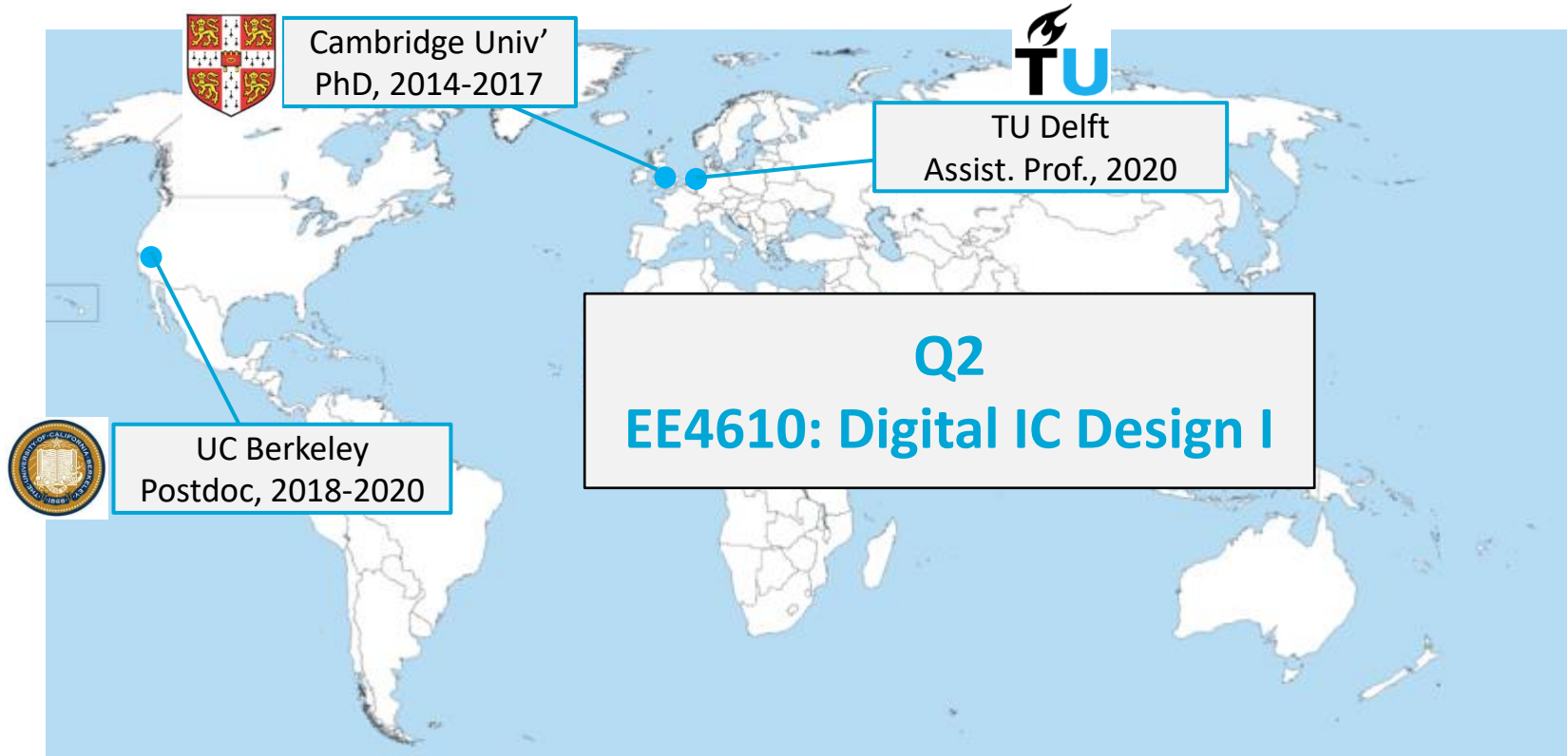
Sijun Du

Electronic Instrumentations Section

Department of Microelectronics

Delft University of Technology

About myself



Electronics around us!



car toy



ceiling fan



chandelier



clock



clothes dryer



mouse



mp3 player



oil-free fryer



piano



coffee maker



computer



drill

**Power Management IC
(PMIC)**



pressure cooker



printer



dishwasher



doorbell camera



DVD player



earphones



projector



radiator



radio



reading lamp



refrigerator



electric frying pan



electric grill



electric guitar



electric pencil sharpener



remote control



rice cooker



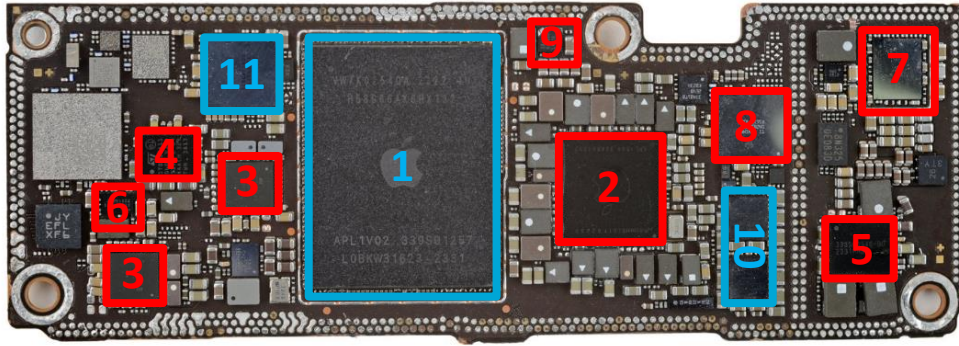
safe



robotic vacuum cleaner

PMICs in our life

Main PCB of Apple iPhone 15 Pro Max



Source: iFixit

1. Apple A17 Pro processor
2. Apple APL109A/338S01022 power management
3. STMicroelectronics STCPM1A3 power management
4. STMicroelectronics STB605A11 power management
5. Apple 338S00946-B0 power management
6. Apple 338S00616 power management
7. Broadcom BCM59365EA1IUBG wireless power receiver
8. Texas Instruments SN2012017 battery charger
9. Texas Instruments TPS61280H battery front-end DC-DC converter
10. Apple 338S00537 audio amplifier
11. Apple 338S00739 CODEC

Internet of Things (IoT)



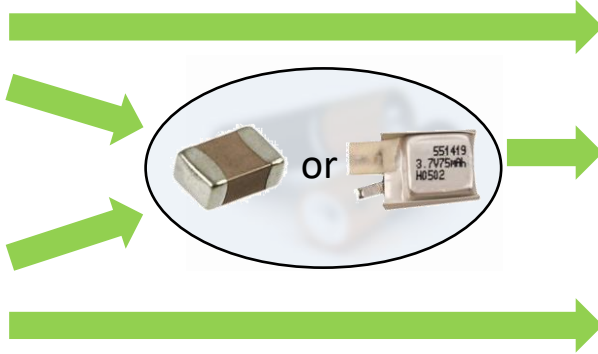
Energy harvesting



Energy harvesting



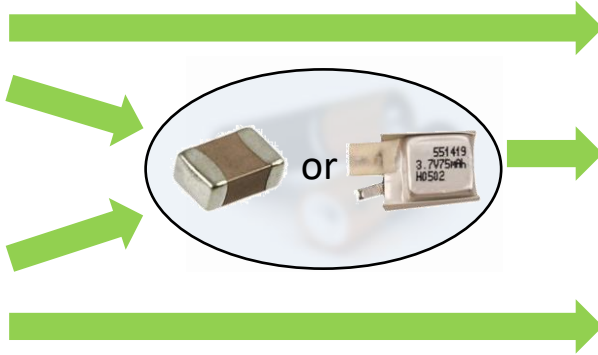
Wireless power
transfer



Energy harvesting



Wireless power
transfer



DC/DC



1.8V
3.3V
5V



Energy harvesting



Wireless power transfer



DC/DC



Current problems:

- External components
- Energy efficiency
- Startup issues

Energy harvesting



Wireless power transfer



DC/DC



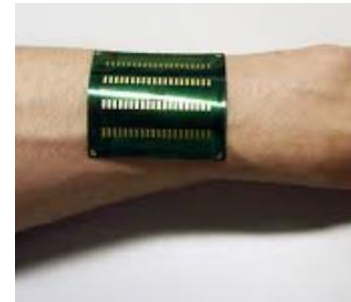
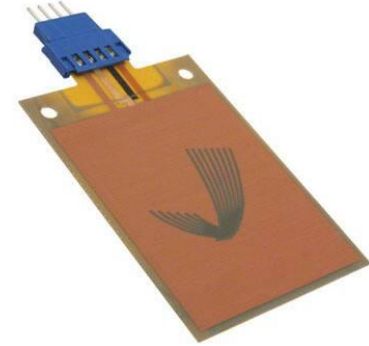
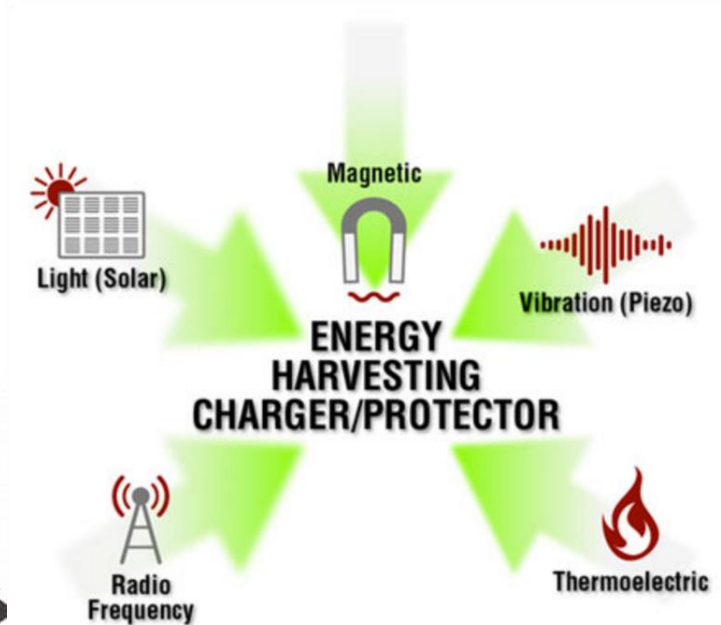
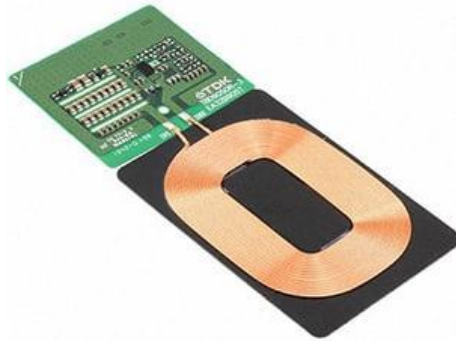
Current problems:

- External components
- Energy efficiency
- Startup issues

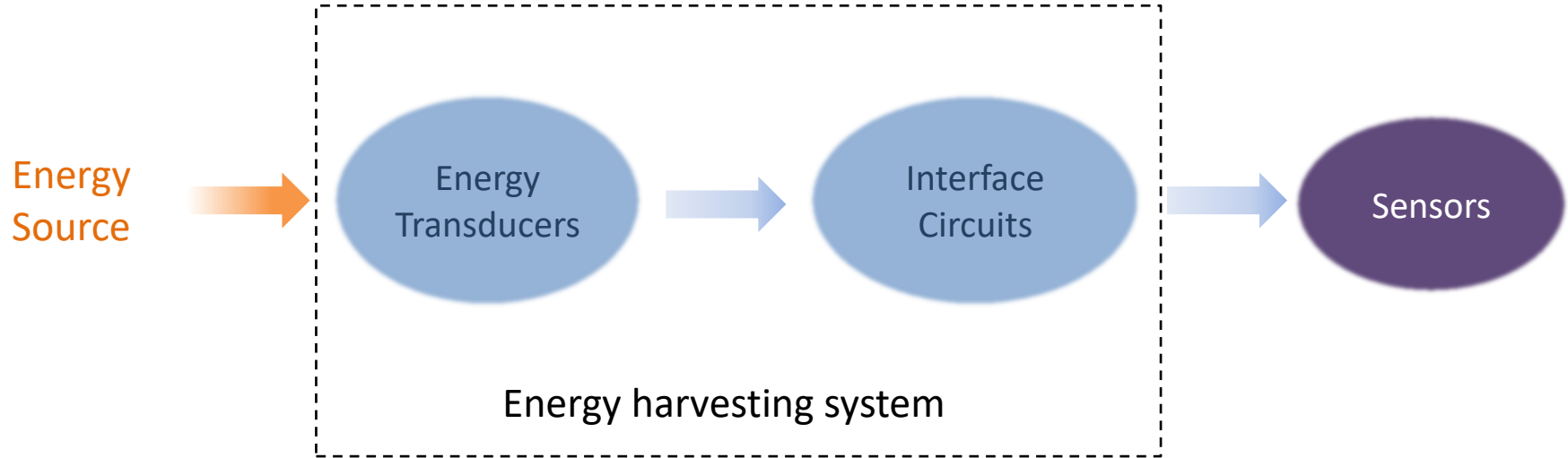
About this lecture

- Focus: End-to-End energy harvesting systems
- In this lecture, you will learn:
 - ✓ Types of energy harvesters
 - ✓ Energy harvester dynamics
 - ✓ Passive rectification circuit design
 - ✓ Active rectification circuit design
 - ✓ Harvested power analysis

Energy harvesting



Energy harvesting systems



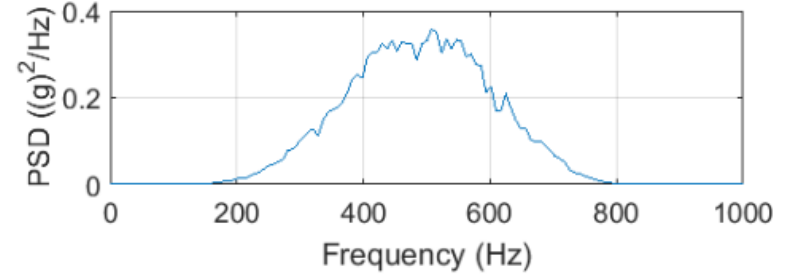
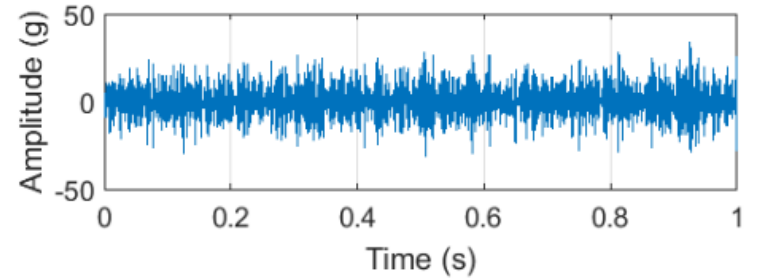
Resonance frequency



City Tram

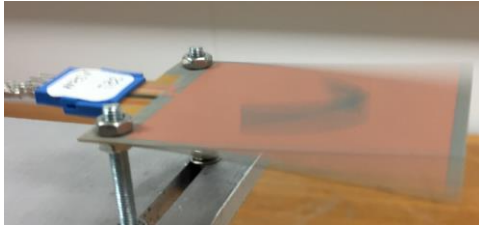


Bridge



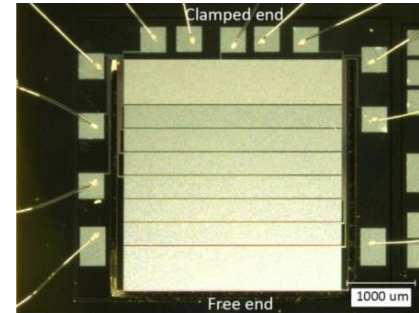
Piezoelectric transducers

Macroscopic
PT



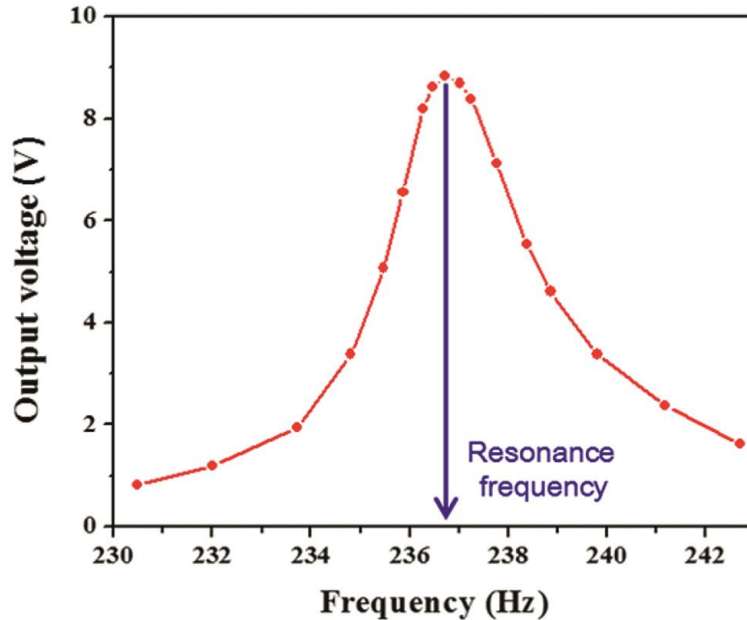
Frequency ~ 200 Hz

MEMS PT



Frequency ~ 500 Hz

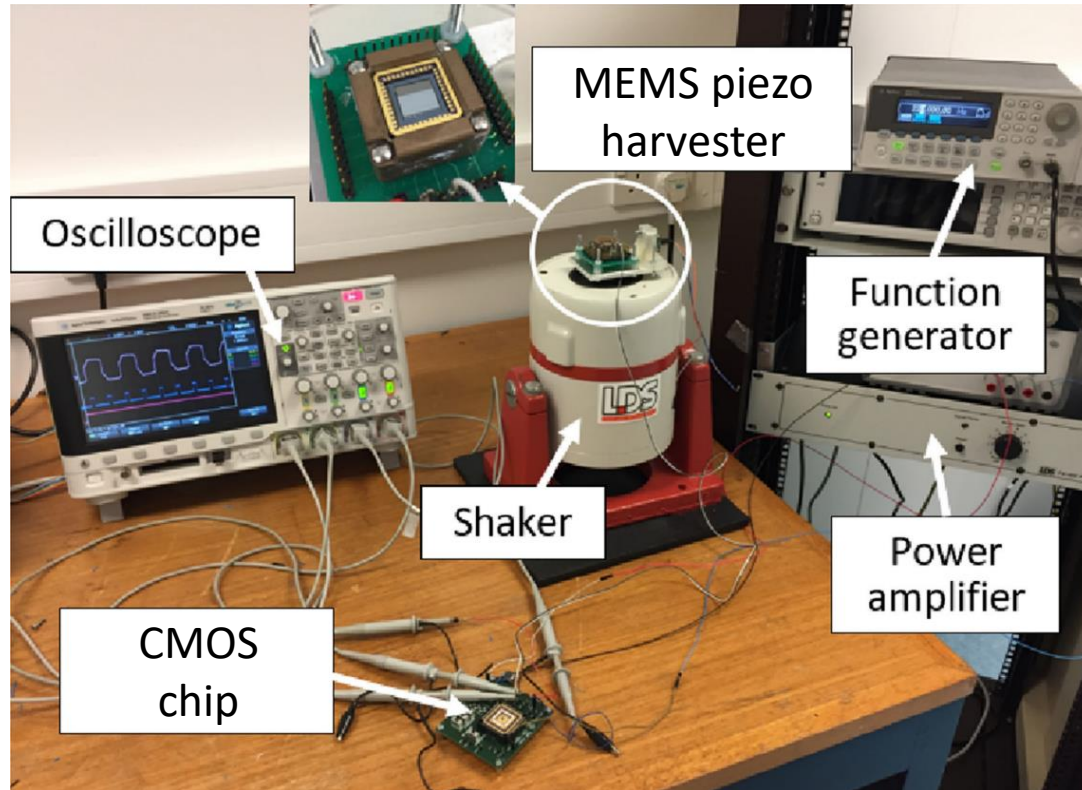
Cantilever designs



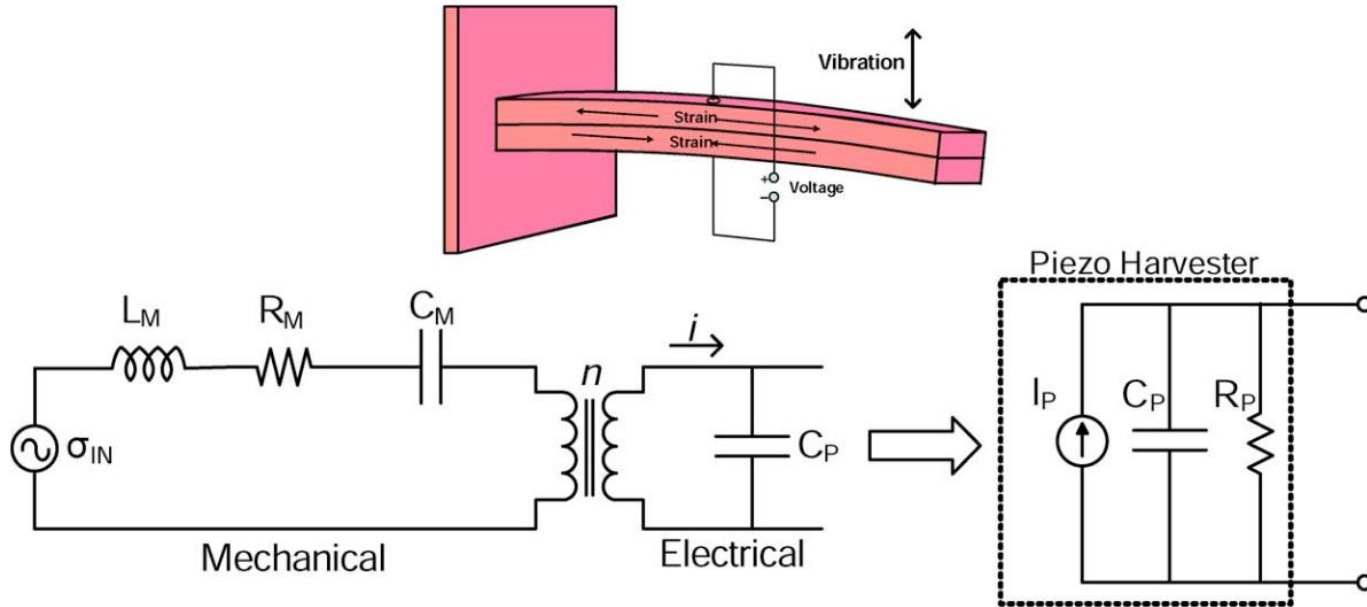
Design considerations:

- Resonance frequency
- Half-power bandwidth
- Acceleration tolerance
- Size
- Operating temperature

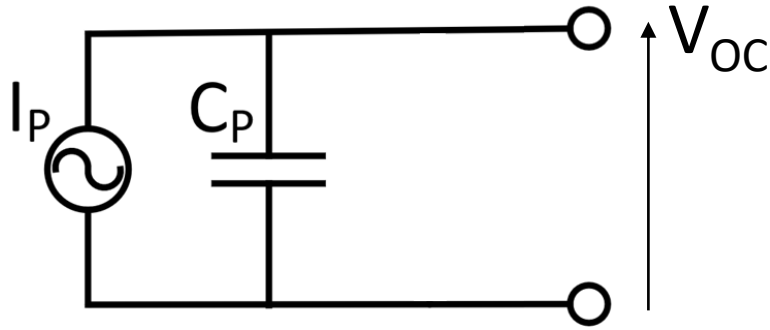
Testing



Simplified model

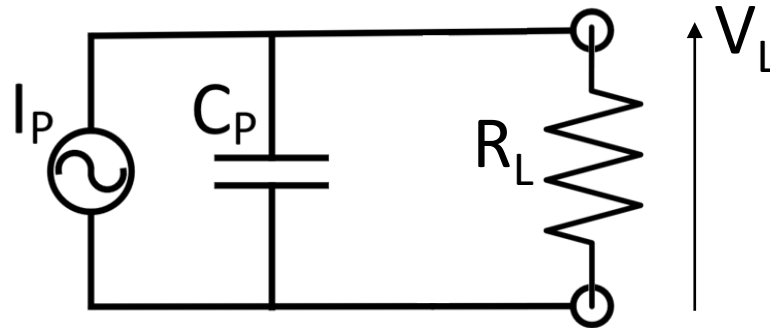


Raw output power



Matched impedance:

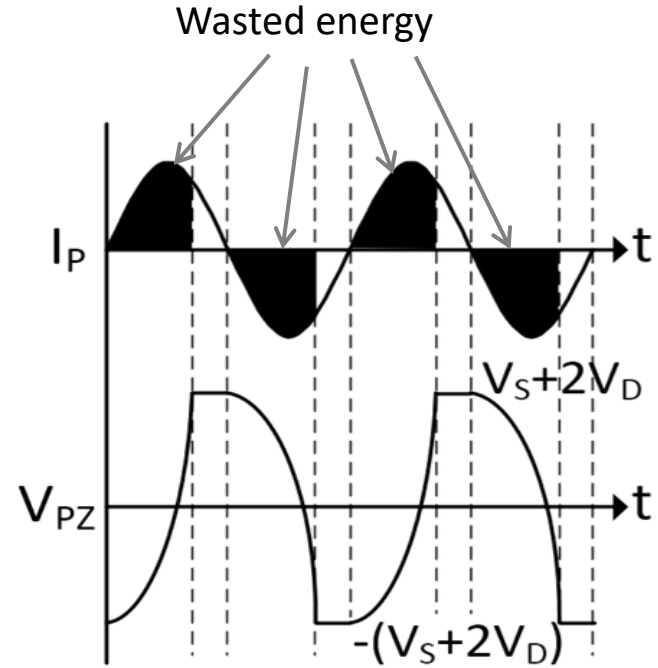
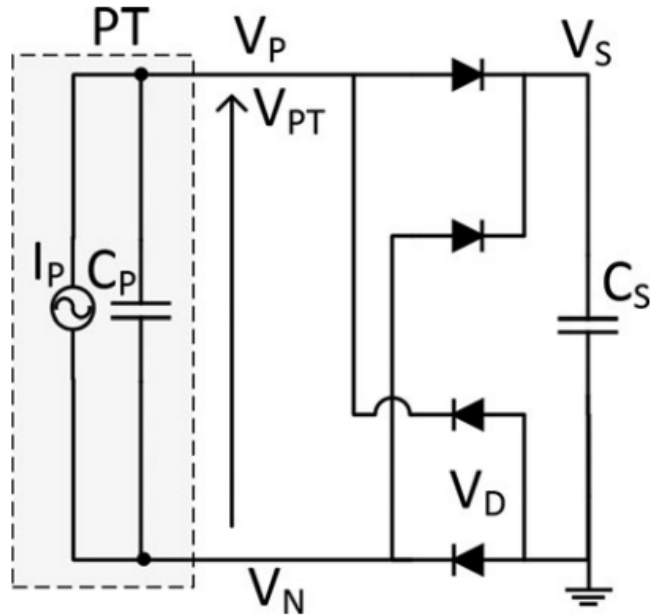
$$R_L = \frac{1}{2\pi f C_P}$$



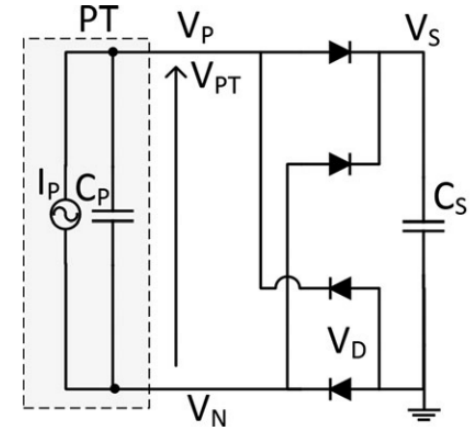
Impedance-matched output power:

$$P_L = \frac{V_L^2}{R_L} = \frac{\left(\frac{V_{OC}}{2}\right)^2}{R_L} = \frac{\pi}{2} f C_P V_{OC}^2$$

Full bridge rectifier



Full bridge rectifier

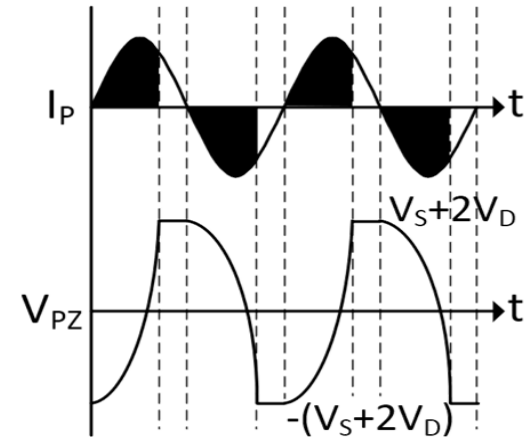


Charge generated in a half period: $Q_{\frac{T}{2}} = \int_0^{\frac{T}{2}} I_0 \sin \omega t \, dt = \frac{2I_0}{\omega}$

Open-circuit voltage: $V_{OC} = \frac{1}{2} \frac{Q_{\frac{T}{2}}}{C_P} = \frac{I_0}{\omega C_P}$

Wasted charge in a half period: $Q_w = 2(V_S + 2V_D)C_P$

Charge flowing into C_S : $Q_S = Q_{\frac{T}{2}} - Q_w = 2V_{OC}C_P - 2(V_S + 2V_D)C_P$
 $= 2C_P(V_{OC} - V_S - 2V_D)$



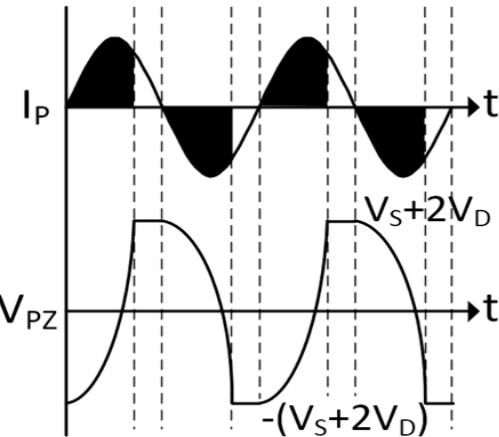
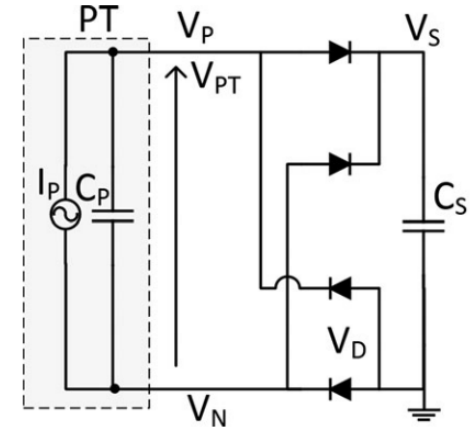
Full bridge rectifier

Charge flowing into C_S : $Q_S = 2C_P(V_{OC} - V_S - 2V_D)$

$$\begin{aligned} \text{Increased energy in } C_S \text{ is: } E_S &= E_{S(t)} - E_{S(t-\frac{T}{2})} \\ &= \frac{1}{2} C_P V_{S(t)}^2 - \frac{1}{2} C_P V_{S(t-\frac{T}{2})}^2 \\ &= \frac{1}{2} C_P (V_{S(t)} + V_{S(t-\frac{T}{2})})(V_{S(t)} - V_{S(t-\frac{T}{2})}) \end{aligned}$$

Assuming voltage increase in C_S in $\frac{T}{2}$ is very small

$$\text{Increased energy in } C_S \text{ is: } E_S \approx C_P V_S \frac{Q_S}{C_P} = 2C_P V_S (V_{OC} - V_S - 2V_D)$$



Full bridge rectifier

Increased energy in C_S is: $E_S = 2C_P V_S (V_{OC} - V_S - 2V_D)$

Power at this V_S in C_S is: $P_S = \frac{E_S}{\frac{T}{2}} = 4C_P V_S f_P (V_{OC} - V_S - 2V_D)$

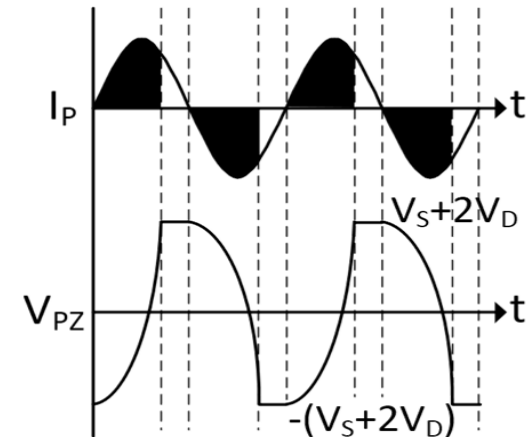
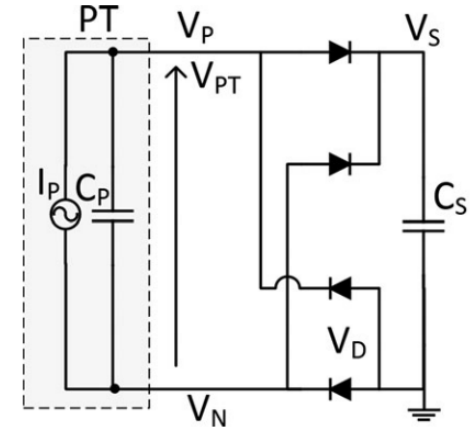
The peak power is: $P_{S(max)} = 4C_P f_P \left(\frac{V_{OC}}{2} - V_D \right)^2$

Optimal V_S for the peak power is: $V_{S(opt)} = \frac{V_{OC}}{2} - V_D$

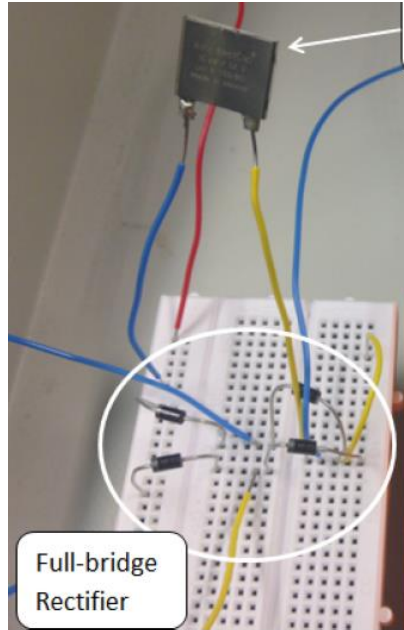
Assuming $V_D \approx 0 \text{ V}$

The peak power is: $P_{S(max)} = C_P f_P V_{OC}^2$

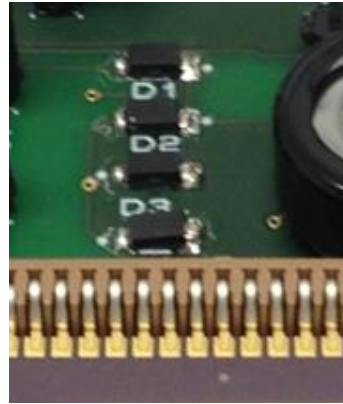
Corresponding to $V_{S(opt)} = \frac{V_{OC}}{2}$



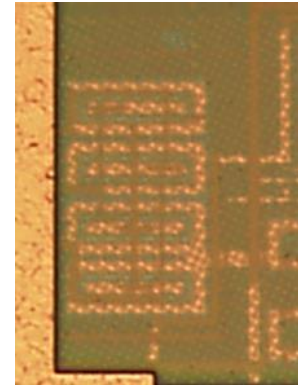
Full bridge rectifier



Breadboard

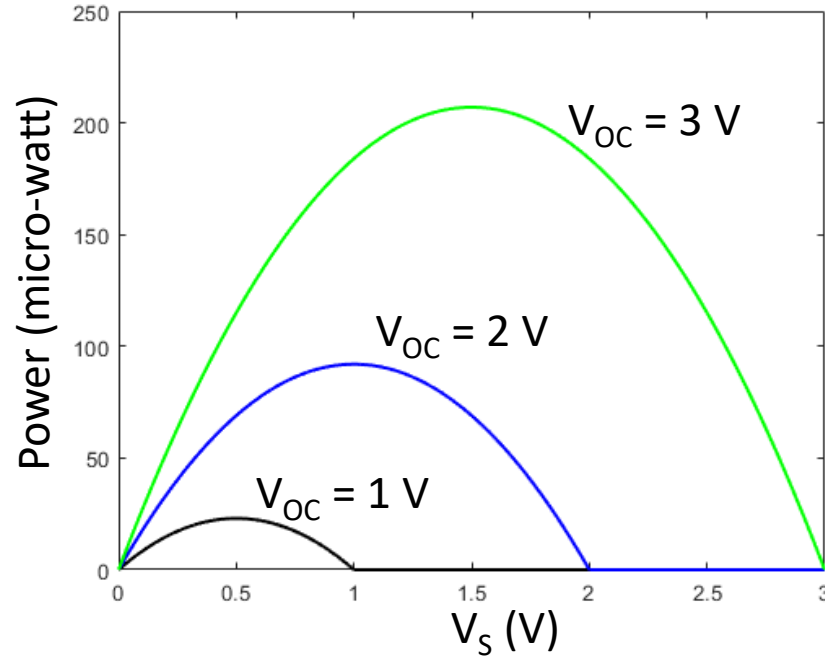


PCB

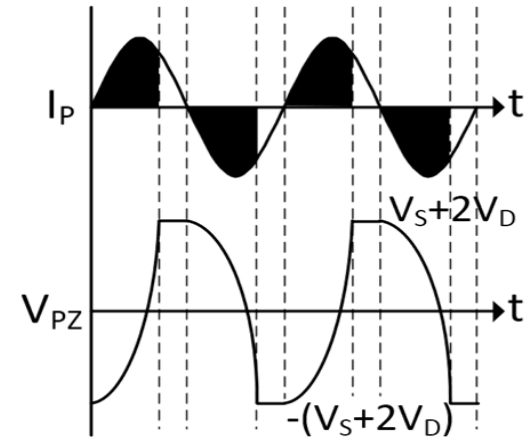
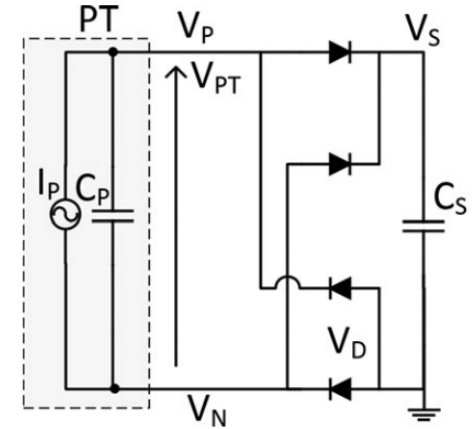


On-chip

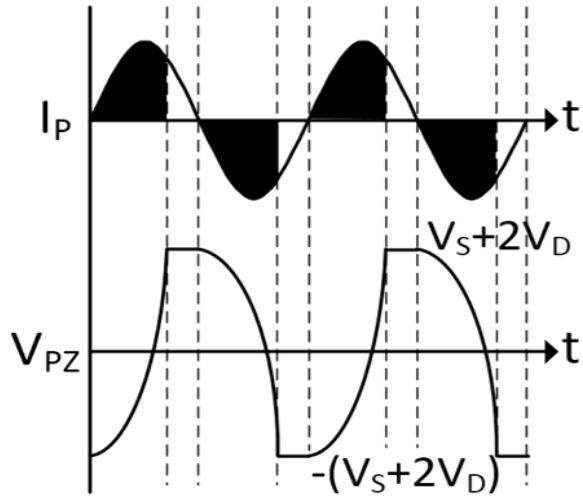
Full bridge rectifier



$$P_S = \frac{E_S}{T} = 4C_P V_S f_P (V_{OC} - V_S - 2V_D)$$

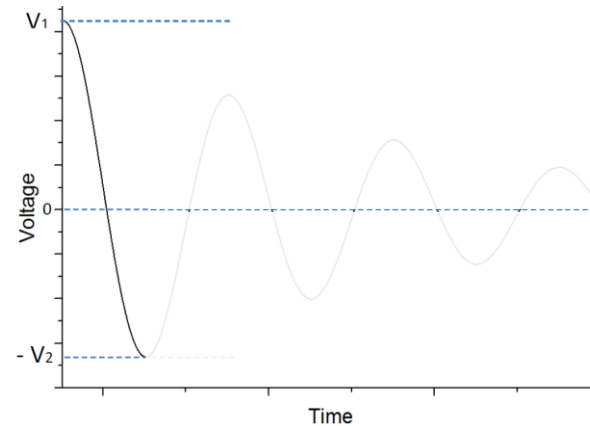
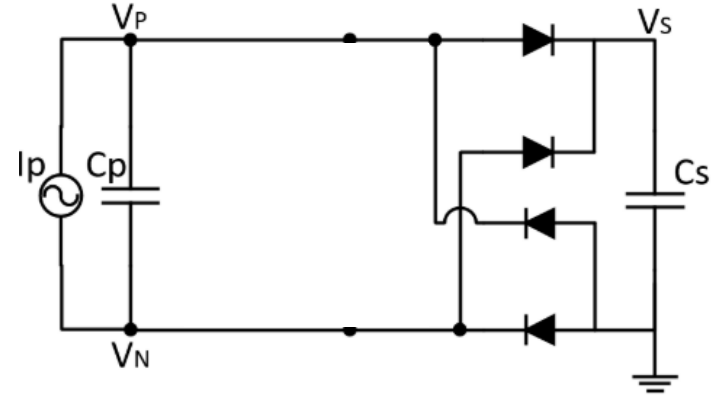


Active rectifier

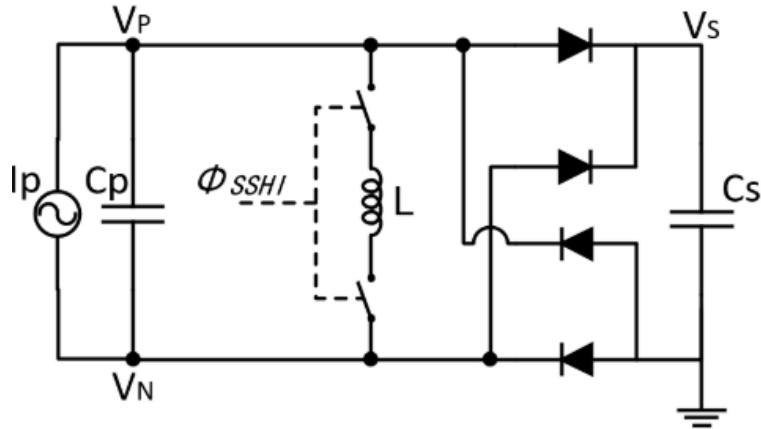


High energy loss

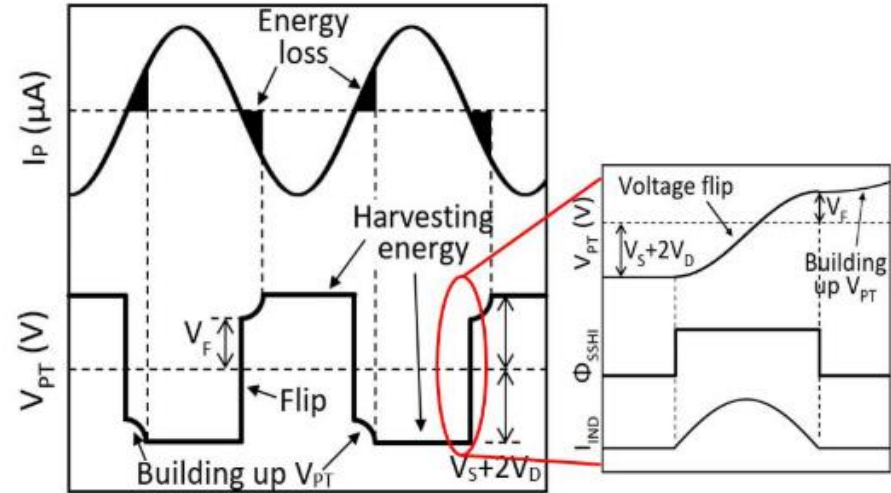
Synchronized switch harvesting on inductor (SSH)



SSH rectifier



SSHI rectifier



Waveform

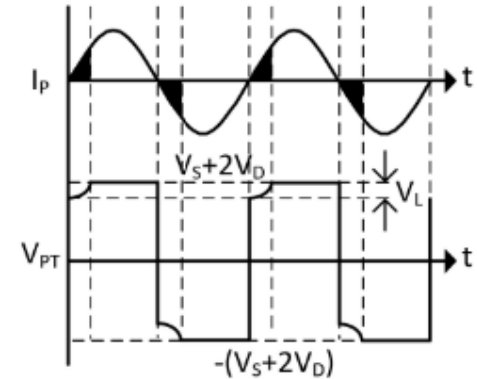
SSH1 performance

Charge generated in a half period: $Q_{\frac{T}{2}} = \int_0^{\frac{T}{2}} I_0 \sin \omega t dt = \frac{2I_0}{\omega}$

Open-circuit voltage: $V_{OC} = \frac{1}{2} \frac{Q_{\frac{T}{2}}}{C_P} = \frac{I_0}{\omega C_P}$

Assuming voltage flip efficiency is η_F

Wasted charge in a half period: $Q_w = (1 - \eta_F)(V_S + 2V_D)C_P$



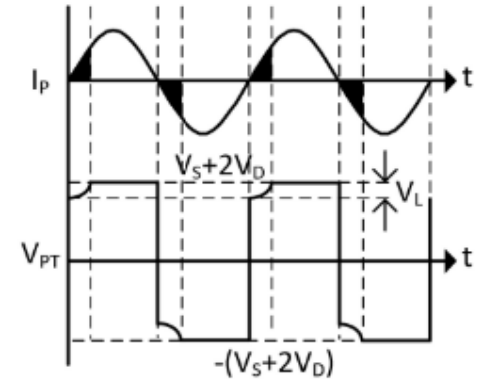
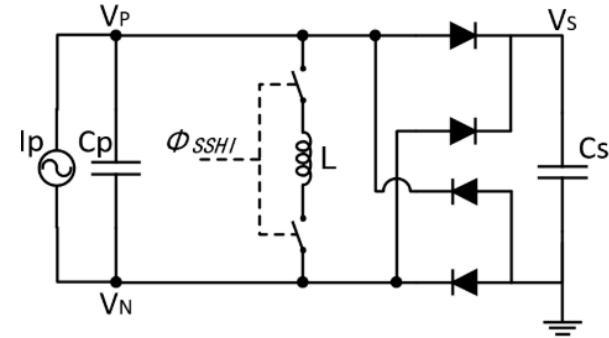
SSH1 performance

Power in C_S is: $P_S = 2f_P C_P V_S (2V_{OC} - (V_S + 2V_D)(1 - \eta_F))$

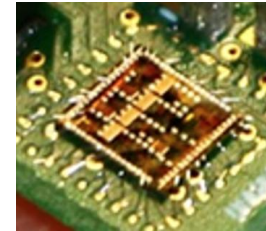
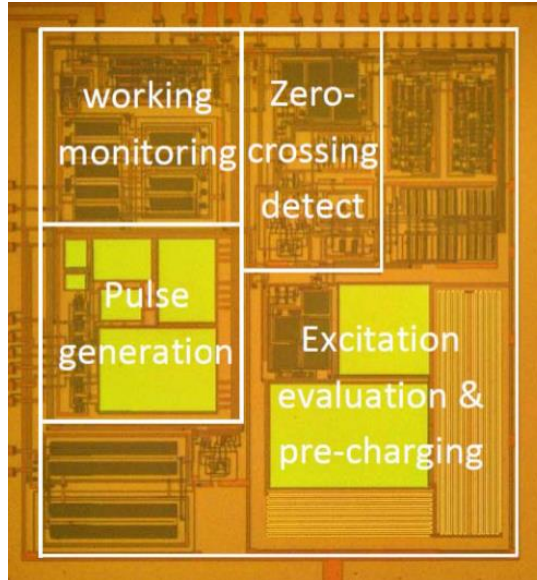
Assuming $V_D \approx 0\text{ V}$

Peak power is: $P_{S(max)} = \frac{2}{1-\eta_F} C_P f_P V_{OC}^2$

Corresponding to $V_{S(opt)} = \frac{V_{OC}}{1-\eta_F}$



SSH circuits



Chip-on-board photo

FBR vs SSHI

FBR

The peak power is: $P_{S(max)} = C_P f_P V_{OC}^2$
when $V_{S(opt)} = \frac{V_{OC}}{2}$

SSHI

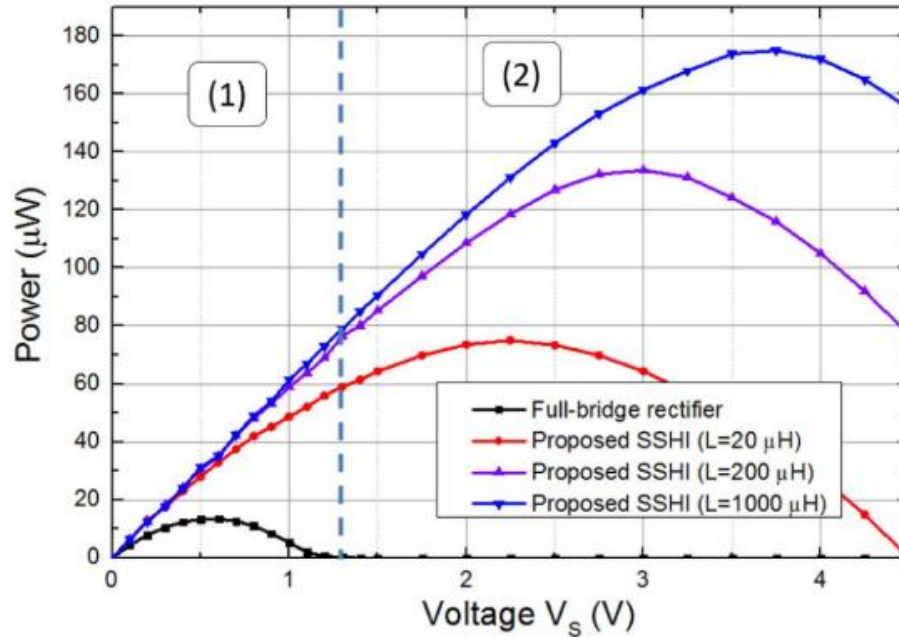
Peak power is: $P_{S(max)} = \frac{2}{1-\eta_F} C_P f_P V_{OC}^2$
when $V_{S(opt)} = \frac{V_{OC}}{1-\eta_F}$

The power enhancement of SSHI over FBR is:

$$\frac{P_{SSHI}}{P_{FBR}} = \frac{2}{1-\eta_F}$$

However, the optimal V_S is also increased by the same factor

FBR vs SSHI



$$\frac{P_{SSHI}}{P_{FBR}} = \frac{2}{1 - \eta_F}$$

Larger inductor means larger voltage flip efficiency: η_F

Inductor?

Most commonly used inductor sizes:

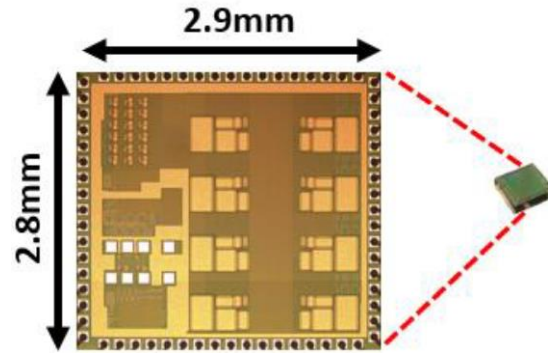
22 μH $\rightarrow \eta_F \approx 40\%$

100 μH $\rightarrow \eta_F \approx 50\%$

220 μH $\rightarrow \eta_F \approx 60\%$

1 mH $\rightarrow \eta_F \approx 70\%$

10 mH $\rightarrow \eta_F \approx 80\%$



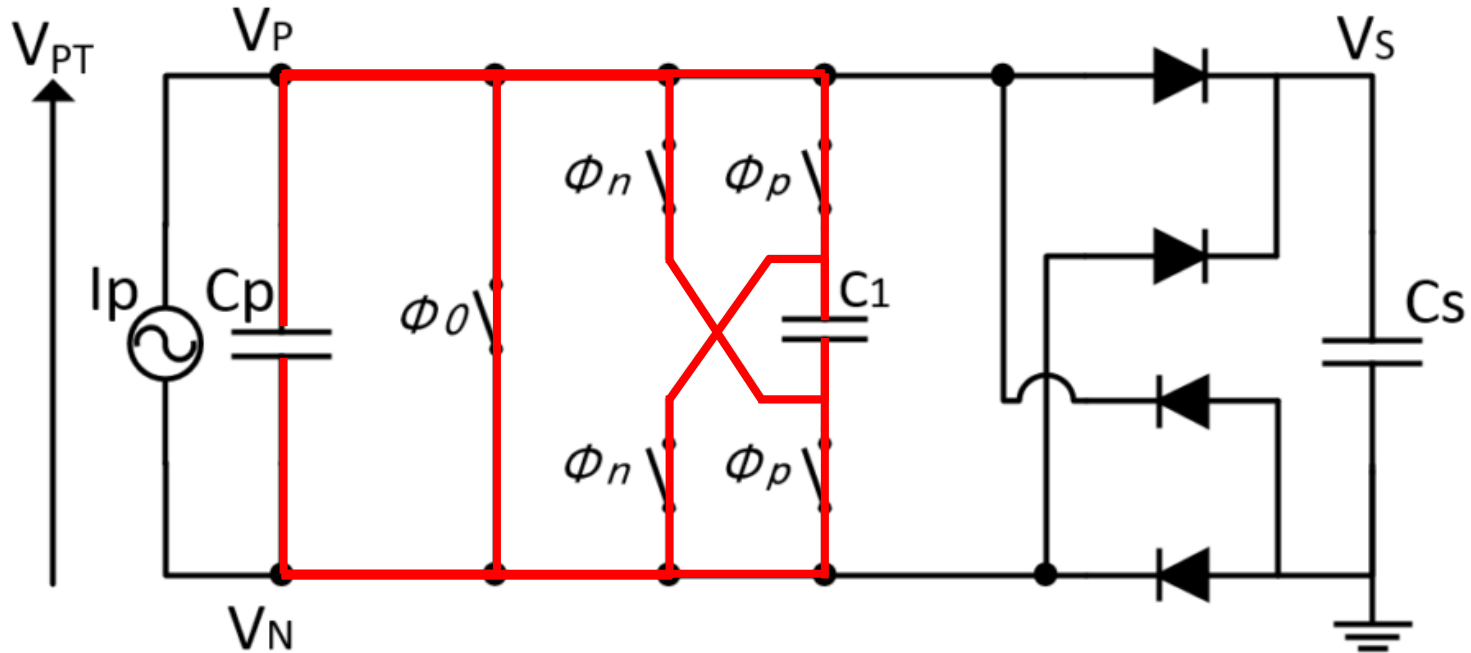
Energy harvesting chip



10mH inductor

How about capacitors?

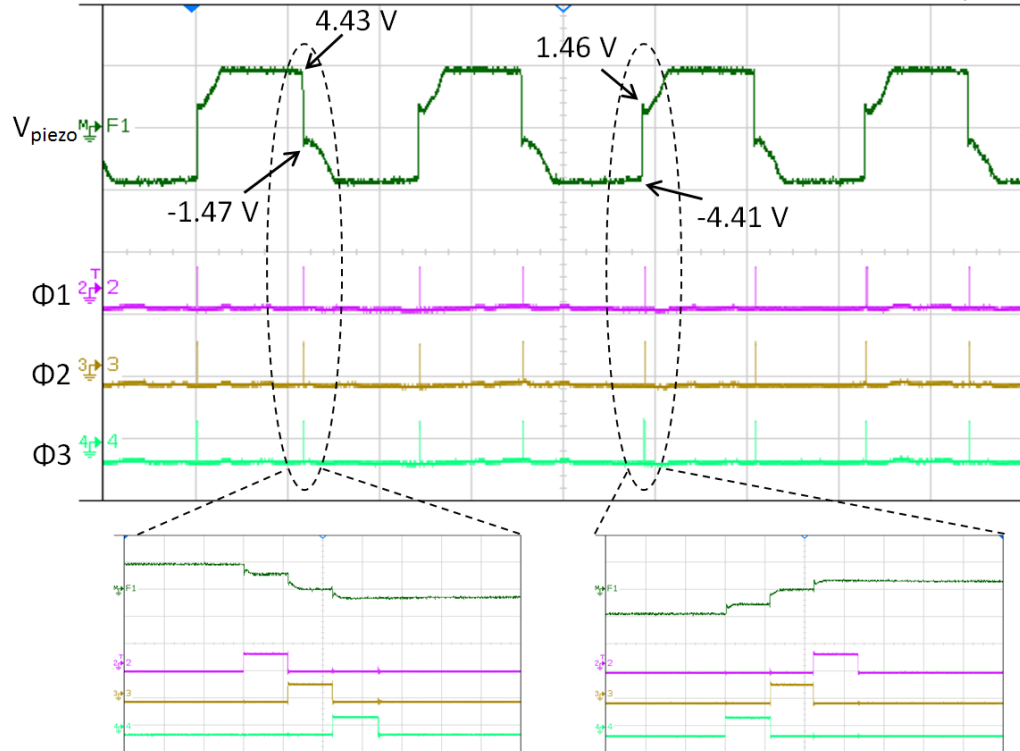
Synchronized switch harvesting on capacitor (SSHC)



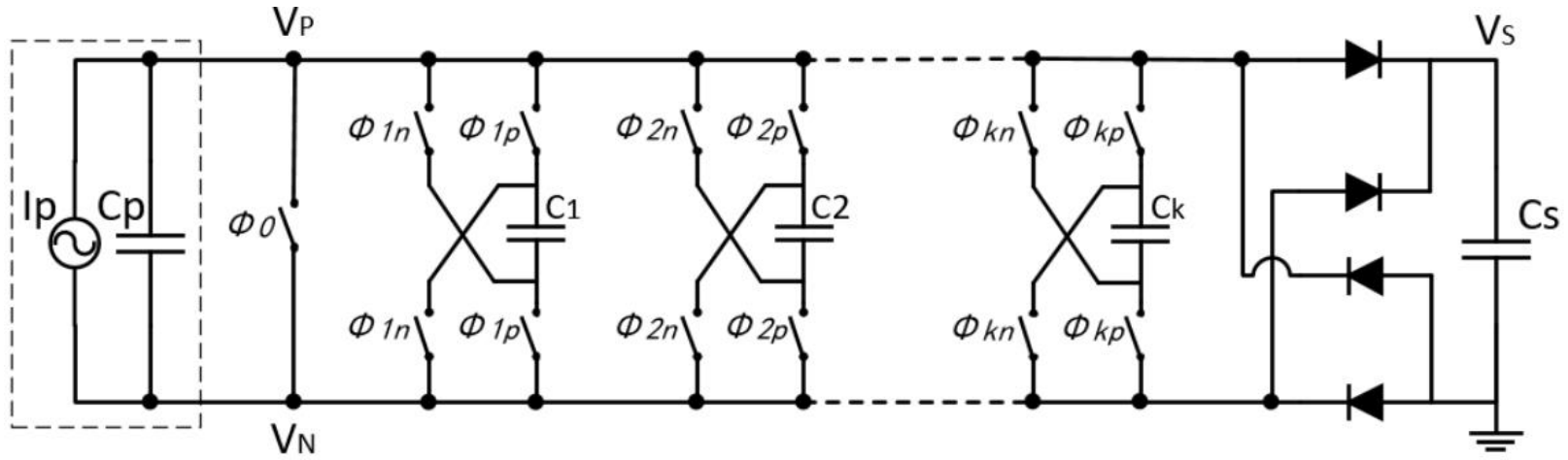
SSHRC rectifier

Using 1 flying cap:

- 3 switching phases
- $\eta_F = \frac{1}{3}$ if $C_1 = C_p$



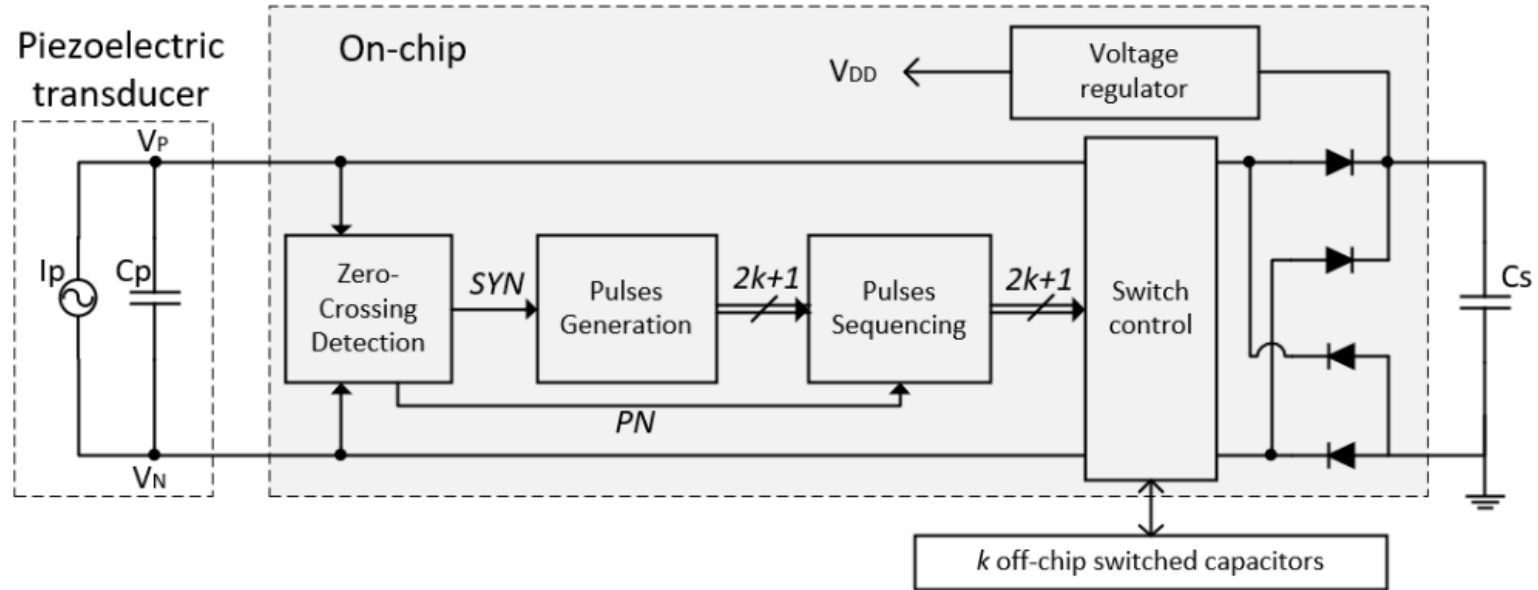
SSHRC rectifier



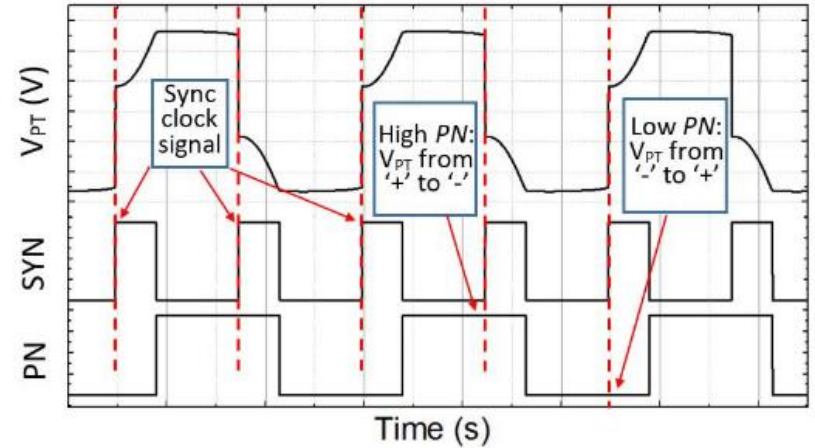
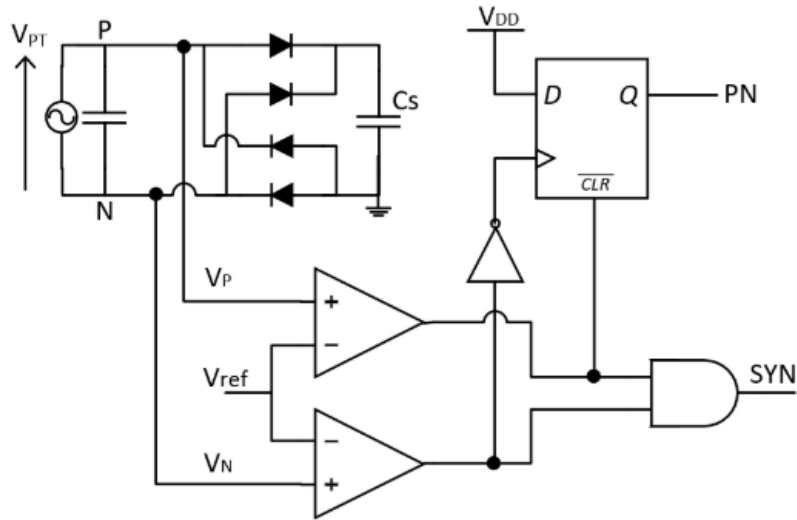
- More stages for more charge dumping per cycle
- Larger C_k for higher flip efficiency
- Peak power is: $P_{S(max)} = \frac{2}{1-\eta_F} C_P f_P V_{OC}^2$

$$\text{Corresponding to } V_{S(opt)} = \frac{V_{OC}}{1-\eta_F}$$

SSHRC rectifier architecture

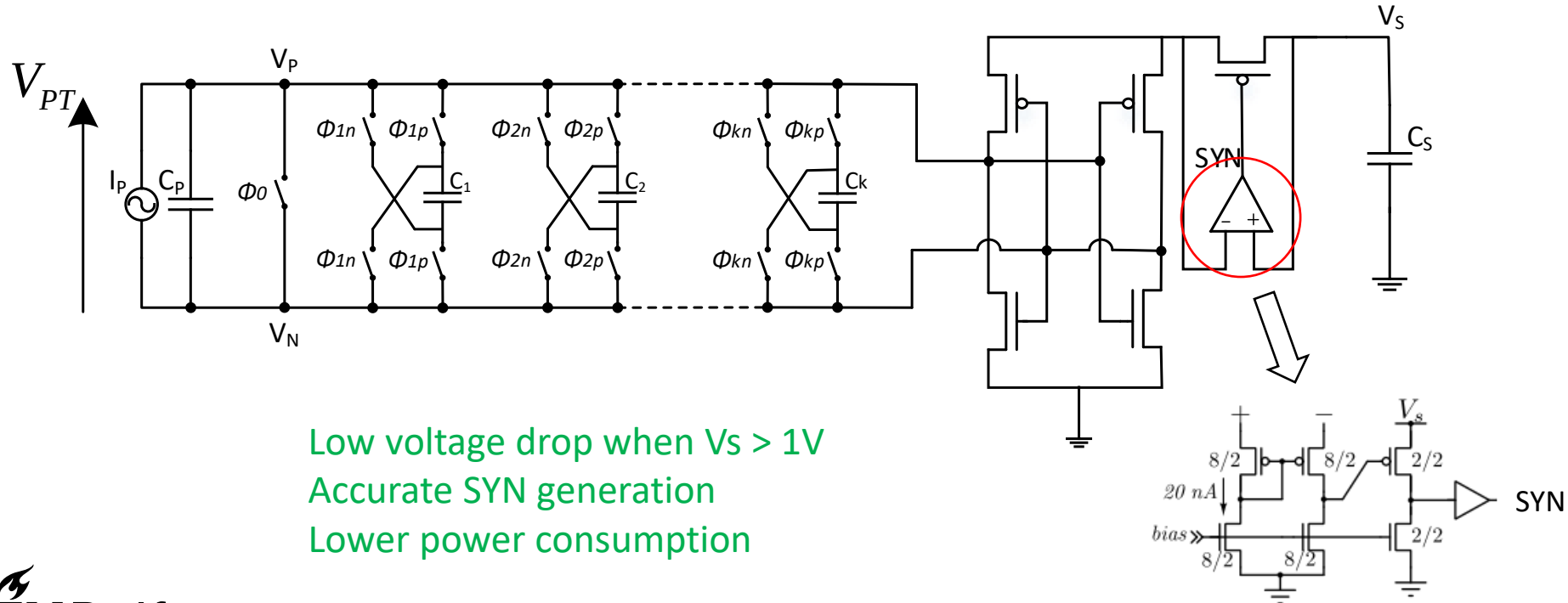


Zero-crossing detection

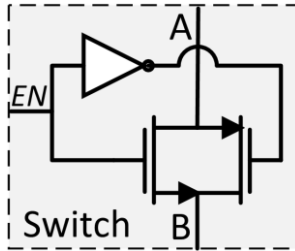


Drawback: Need a reference voltage V_{ref}

Zero-crossing detection



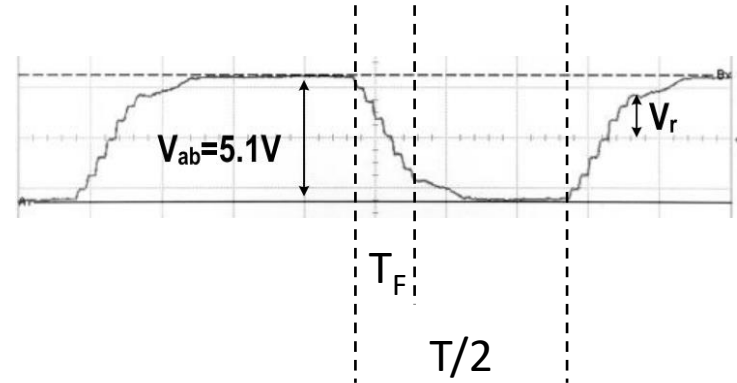
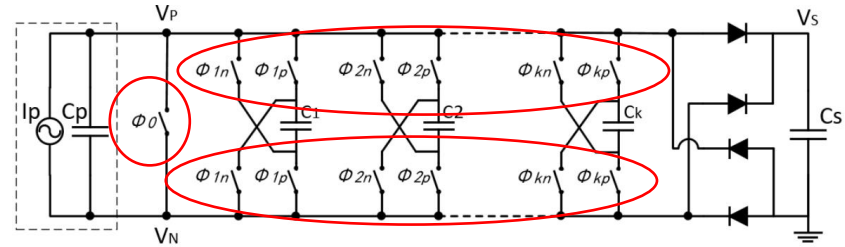
Switch design



Increasing W/L will result

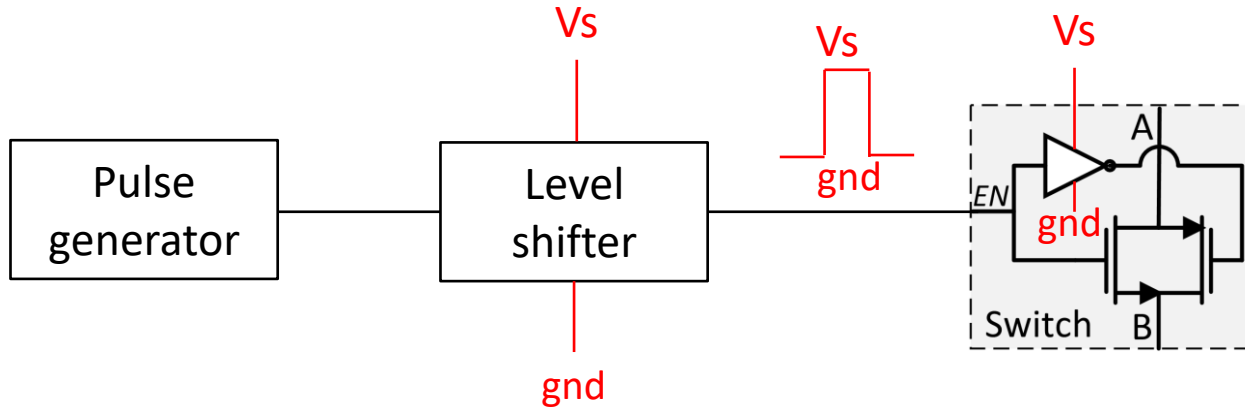
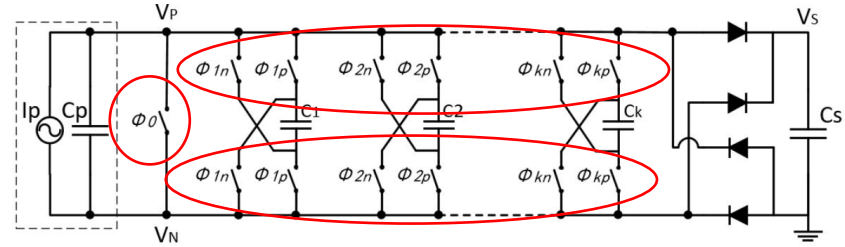
- Shorter flip time
- Higher switching loss

A trade-off between output power and power consumption

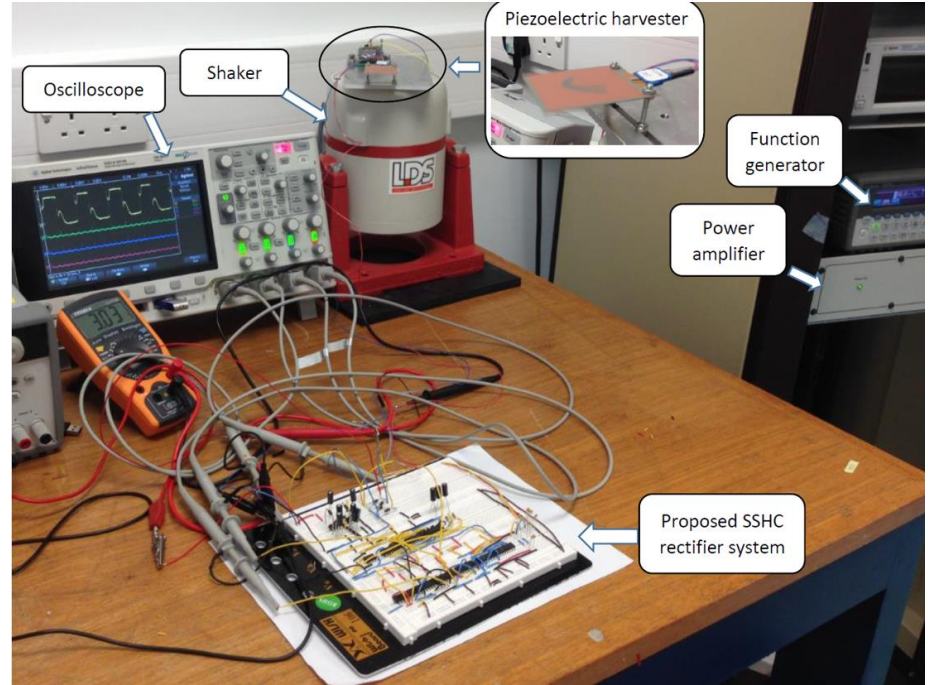
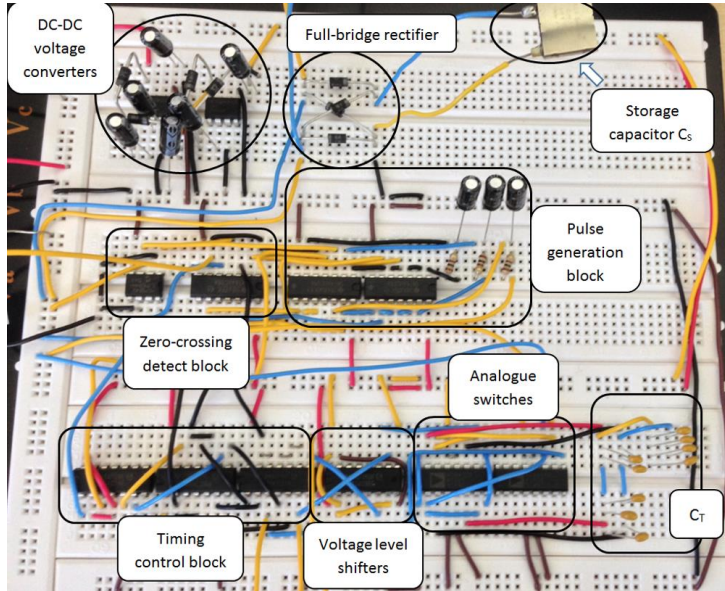


We want $T_F \ll T/2$ to minimize energy loss
 T_F depends on the RC time constant of each loop

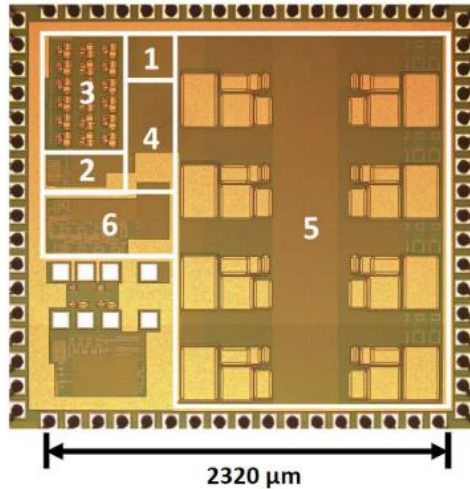
Switch design



Breadboard 1-stage SSHC

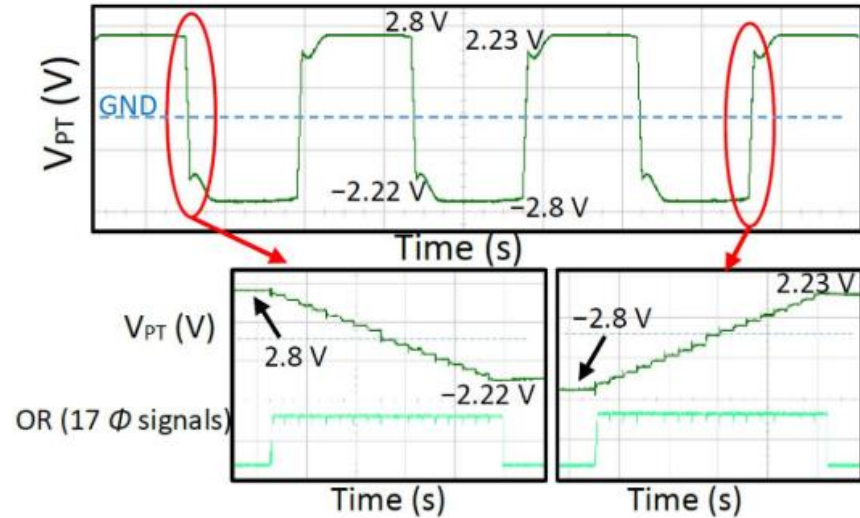


CMOS 8-stage SSHC



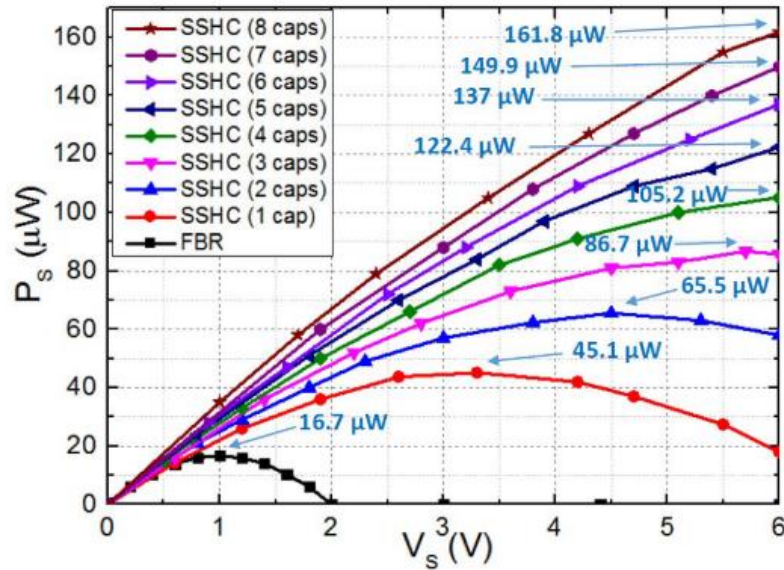
180nm HV, ISSCC'18

World's first inductor-less architecture

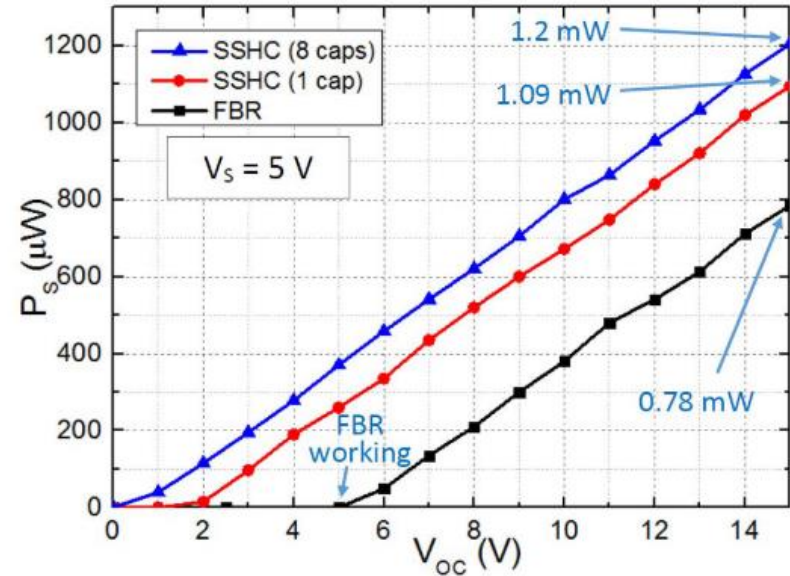


- All 8 flying capacitors enabled
- 17 switching phases in total
- ~80% flip efficiency

CMOS 8-stage SSHC

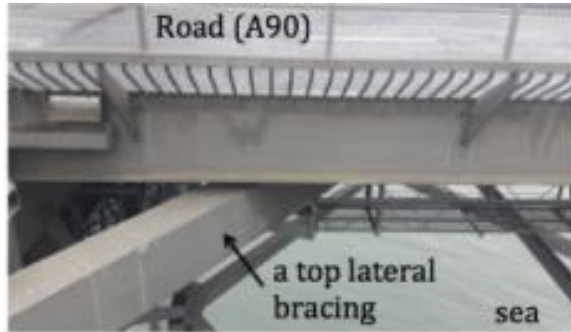


Fixed $V_{OC} = 2.5\text{V}$
In a range of rectified voltage



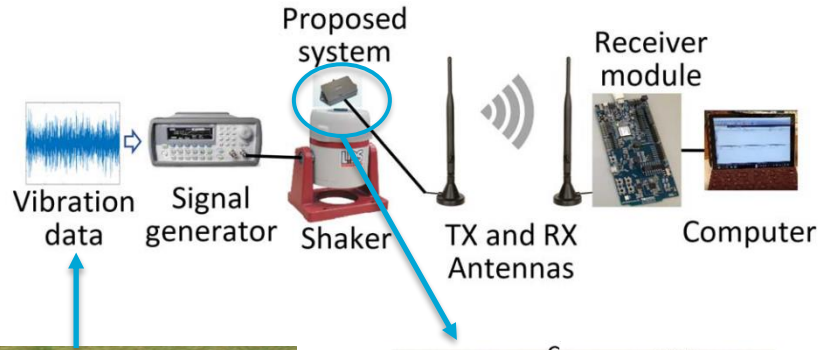
Fixed rectified voltage $V_s = 5\text{V}$
In a range of excitation amplitude

Real-world implementation

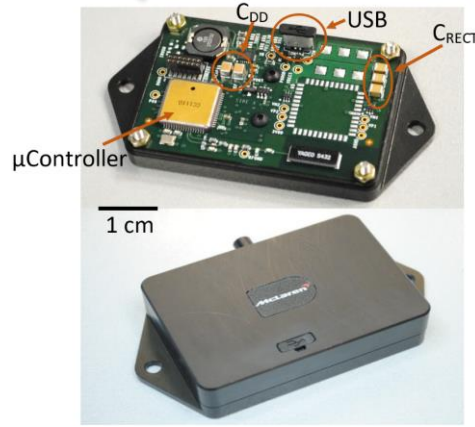


Self-powered wireless sensor under a bridge in Scotland (2017)

Real-world implementation: F-1 car



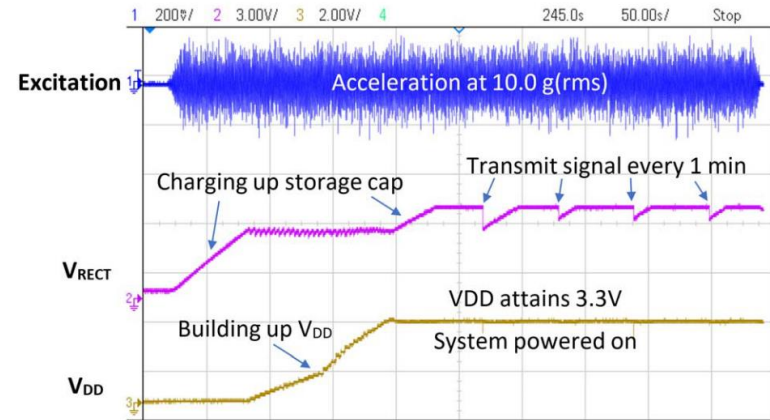
Realtime data
collected from a
McLaren F-1 car



TRANSDUCERS'19

System features:

- Fully self-sustained (battery-less)
- Cold start (all caps fully discharged)
- Custom MEMS energy harvesters with optimized interface circuits
- Fully functional prototype, with MCU, DSP, sensors, TRX, etc.



Homework

#1: Design a fully bridge rectifier

#2: Design an active rectifier with only a switch to clear residual charge

Deliverable: A report giving the schematic, waveform, output power, and some explanations.

Homework #1

Designing a full-bridge rectifier

PT parameters:

$$I_P = I_0 * \sin 2\pi f$$

Where $I_0 = 50 \mu\text{A}$

$f = 200 \text{ Hz}$

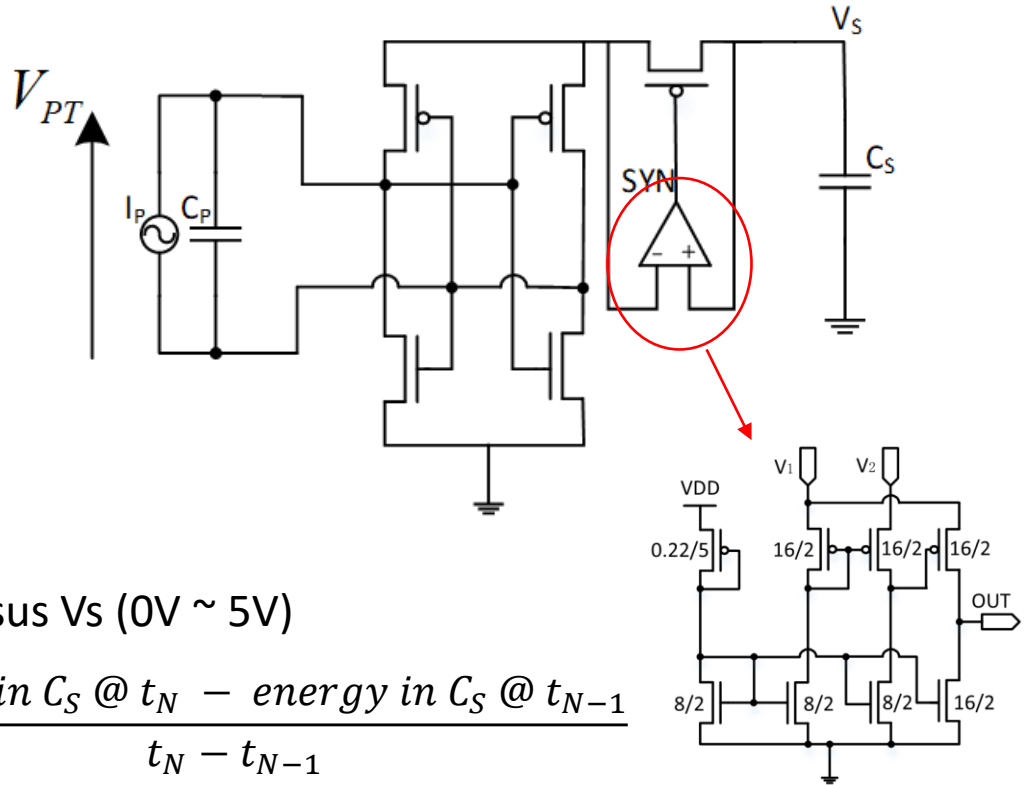
$C_P = 10 \text{ nF}$

C_S between $1 \mu\text{F}$ and $100 \mu\text{F}$

Deliverables:

- Simulated V_{PT} waveform.
- Simulated figure of Power versus V_S ($0\text{V} \sim 5\text{V}$)

$$\text{Power @ } t_N = \frac{\text{Energy in } C_S \text{ @ } t_N - \text{energy in } C_S \text{ @ } t_{N-1}}{t_N - t_{N-1}}$$



Homework #2

Designing a zero-stage SSHC rectifier

PT parameters:

$$I_P = I_0 * \sin 2\pi f$$

Where $I_0 = 50 \mu\text{A}$

$f = 200 \text{ Hz}$

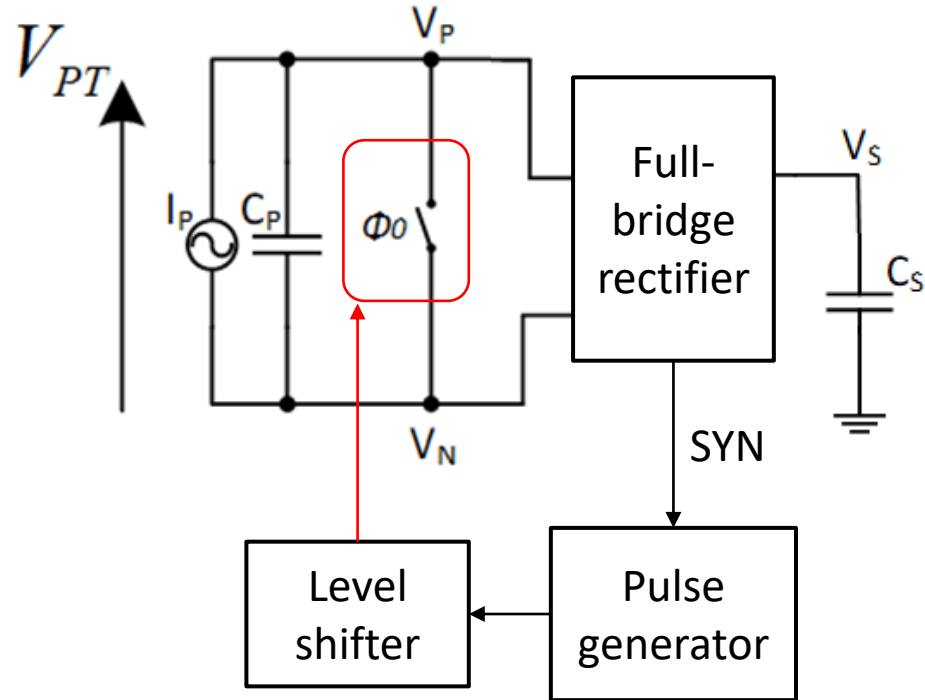
$C_P = 10 \text{ nF}$

C_S between $1 \mu\text{F}$ and $100 \mu\text{F}$

$V_{DD} = 1.5 \text{ V}$

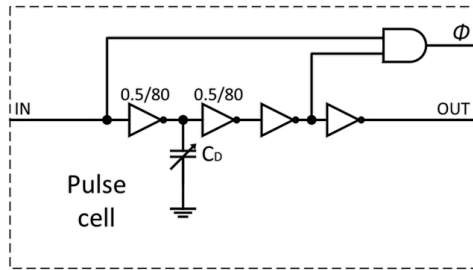
Deliverables:

- Simulated V_{PT} waveform showing voltage flipping
- Simulated figure of Power versus V_S ($0\text{V} \sim 5\text{V}$)
- Compare your power figure with HW #1.

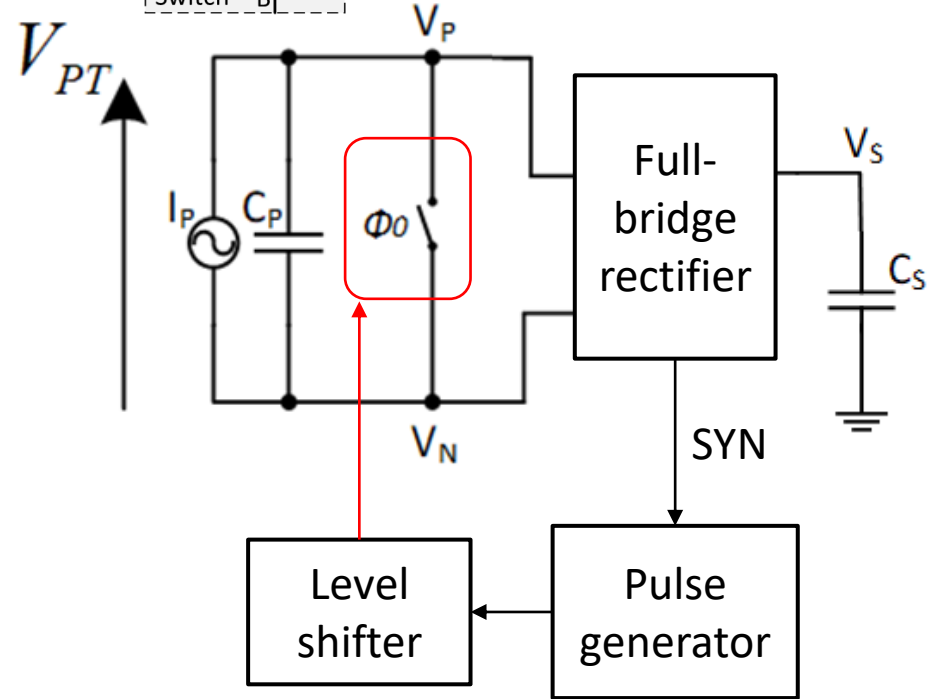
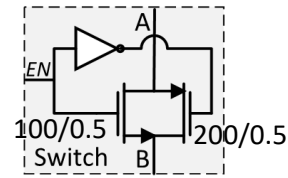
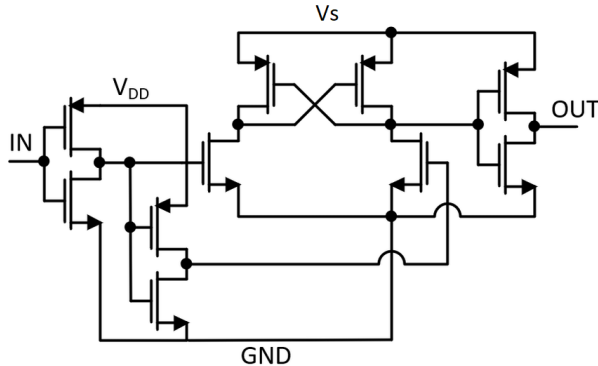


Homework #2

Pulse generator



Level shifter



TA of this homework



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Reference

- No reference textbook.
- Reference papers (already on Bright Space)

- [1] P. Shashank et al., "A Review on Piezoelectric Energy Harvesting: Materials, Methods, and Circuits," (in English), Energy Harvesting and Systems, vol. 4, no. 1, pp. 3-39, 2019, doi: <https://doi.org/10.1515/ehs-2016-0028>
- [2] Y. K. Ramadass and A. P. Chandrakasan, "An Efficient Piezoelectric Energy Harvesting Interface Circuit Using a Bias-Flip Rectifier and Shared Inductor," IEEE Journal of Solid-State Circuits, vol. 45, no. 1, pp. 189-204, 2010, doi: 10.1109/JSSC.2009.2034442
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